TAIST TokyoTech

F03- Research Progress Report.

TAIST-Tokyo Tech

Research Progress Report
For Living allowance scholarship

Academic Year......

☑ 1st ☐ 2nd ☐ 3rd ☐ 4th ☐ 5th ☐ 6th ☐ 7th ☐ 8th Research Progress Report

Part 1: Student information

Name Nontanan Surname Sommat

Student ID 68070703807 University King Mongkut's University of Technology Thonburi

Program:

☑ Dept. of Automotive Engineering (A2TE)

☐ Dept. of Information and Communication Technology for Embedded System (AIoT)

☐ Dept. of Sustainable Energy and Resources Engineering

Part 2: University Advisor information

Advisor: Asst. Prof. Dr. Danai Surname Phaoharuhansa

Part 3: NSTDA Researcher Information

NSTDA researcher: .... Surname ....

Research Center: ☐ NECTEC ☐ MTEC ☐ BIOTEC ☐ NANOTEC

Laboratory: ....

Part 4: Thesis/Research Information

Title of thesis/research:

Development of an Adaptive Path Tracking Control System for a Multi-Trailer Autonomous Tow Truck using Metaheuristic Optimization in CARLA Simulator

* Introduction

The operation of autonomous towing tractors with multiple trailers (Standard N-Trailer) presents significant challenges in trajectory tracking due to complex kinematic constraints, off-tracking effects, and potential instability such as jackknifing. Furthermore, real-world implementations often suffer from physical uncertainties like mechanical wear and steering bias, which degrade the performance of standard controllers. This research proposes an adaptive control framework integrating Nonlinear Model Predictive Control (NMPC) with a Grey Wolf Optimizer (GWO). The GWO functions as an online observer to estimate and compensate for steering bias in real-time, ensuring precise tracking for a tractor with four drawbar trailers.

* Scopes/Objectives

1) To derive and validate the kinematic model of a tractor with four drawbar trailers, specifically accounting for off-axle hitching geometry.

2) To develop a Nonlinear Model Predictive Control (NMPC) system capable of handling multi-body constraints and ensuring stability.

3) To implement the Grey Wolf Optimizer (GWO) as an adaptive estimator to compensate for steering bias and disturbances.

4) To validate the system performance using CARLA Simulator integrated with ROS 2 and Autoware.

* *Write only at the first time of report (or rewrite if there is any change from the previous report)*

F03- Research Progress Report.

Research Plan

1. Research plan: give a brief explanation of each activity/procedure, and report recent progress, completed activity, and future plan)

Phase 1: Development Environment & Middleware Deployment

Setting up the simulation testbed involved installing Ubuntu 22.04 LTS and ROS 2 Humble. The system was configured with NVIDIA Drivers and Autoware Universe to enable hardware acceleration. Afterward, CARLA Simulator 0.9.15 was deployed natively, followed by benchmarking to ensure the workstation can handle multi-agent physics simulations. **Status: [COMPLETED]**

Phase 2: Mathematical Modeling & Kinematic Analysis

Theoretical foundation work focuses on vehicle dynamics, starting with the derivation of state-space kinematic equations for the tractor with four drawbar trailers, explicitly accounting for off-axle hitching geometry. These equations will be validated via an open-loop Python simulation. Following this validation, the tractor_odometry ROS 2 package will be developed to broadcast real-time pose estimations. Simultaneously, a comprehensive study of Metaheuristic Algorithms will be conducted to select the optimal strategy for parameter tuning **Status: [2.1-2.3 COMPLETED].**

Phase 3: Controller Framework Development (NMPC)

Design and implementation of the core control architecture focuses on formulating the NMPC cost function to minimize tracking error while enforcing safety constraints, such as jackknifing prevention. The solver will be implemented using CasADi with Python simulation. To integrate effectively with the autonomous stack, two interface nodes will be developed: planning_interface for trajectory parsing and control_interface for command transmission. Finally, an integration test will verify the communication pipeline between the custom package and Autoware.

Phase 4: Adaptive System Integration (AI)

Implementation of the intelligent adaptation layer entails coding the Grey Wolf Optimizer (GWO) algorithm class structure. Logic for estimating "Steering Bias" from steady-state tracking errors will be developed to function as a disturbance observer. The final step involves integrating this GWO observer with the NMPC prediction model to achieve Closed-Loop Adaptation, enabling the system to compensate for physical uncertainties in real-time.

Phase 5: High-Fidelity Environment & Asset Construction

Creation of realistic simulation assets includes designing dimensionally accurate 3D models of the tractor and trailers using SolidWorks and Blender. Furthermore, HD Maps (.xodr) and Point Clouds (.pcd) will be generated for Autoware localization. These assets will be used for full system integration, merging CARLA, Autoware, and the control package, followed by initial control parameter tuning to establish a stable baseline.

Phase 6: Validation, Analysis & Dissemination

The final phase involves designing experiments and defining metrics to evaluate system performance. The goal is to achieve a success rate of over 95%. Testing will cover complex scenarios, such as sharp cornering and fault handling, to verify robustness. Results will be compiled into the final thesis and prepared for publication.

Table 1 Research Plan

Research Plan (Activities)	1	2	3	4	5	6	7	8	9	10	11	12
1. Development Environment & Middleware												
1.1 Install Ubuntu 22.04 & ROS 2 Humble	C											
1.2 Setup GPU Driver & Install Autoware Universe	C											
1.3 Install CARLA 0.9.15 (Native) & Benchmarking	C											
2. Mathematical Modeling & Metaheuristics												
2.1 Study Kinematics & Derive 4-Trailer Equations	C											
2.2 Simulate Kinematic Model in Python (Validation)	C											
2.3 Develop ROS 2 Package tractor_odometry	C											
2.4 Study Metaheuristics Algorithm	IP											
3. Controller Development (NMPC)												
3.1 Design NMPC cost function & constraints		X										
3.2 Implement Solver (CasADi) & Python Simulation		X										
3.3 Develop planning_interface (Trajectory Bridge)		X										
3.4 Develop control_interface (Cmd Override)		X										
3.5 Integration Test (Custom Pkg vs Autoware)		X	X									
4. Adaptive System Development (AI)												
4.1 Develop Grey Wolf Optimizer (GWO) algorithm			X	X								
4.2 Implement "Steering Bias" estimation logic			X	X								
4.3 Integrate GWO with NMPC (Closed-loop)				X	X							
5. Simulation Environment & Assets												
5.1 Design 3D Models (SolidWorks/Blender)					X	X						
5.2 Create HD Map (.xodr) & Point Cloud (.pcd)						X	X					
5.3 Full System Integration (CARLA+Autoware+AI)							X	X	X			
5.4 Control Parameter Tuning & Optimization									X	X		
6. Testing, Validation & Docs												
6.1 Design Experiment & Evaluation Metrics										X	X	
6.2 Performance Validation (>95%)										X	X	
6.3 Scenario Testing & Analysis											X	X
6.4 Thesis Writing											X	X

Note: C is Completed (Finished), IP is In Progress, X is Future Plan

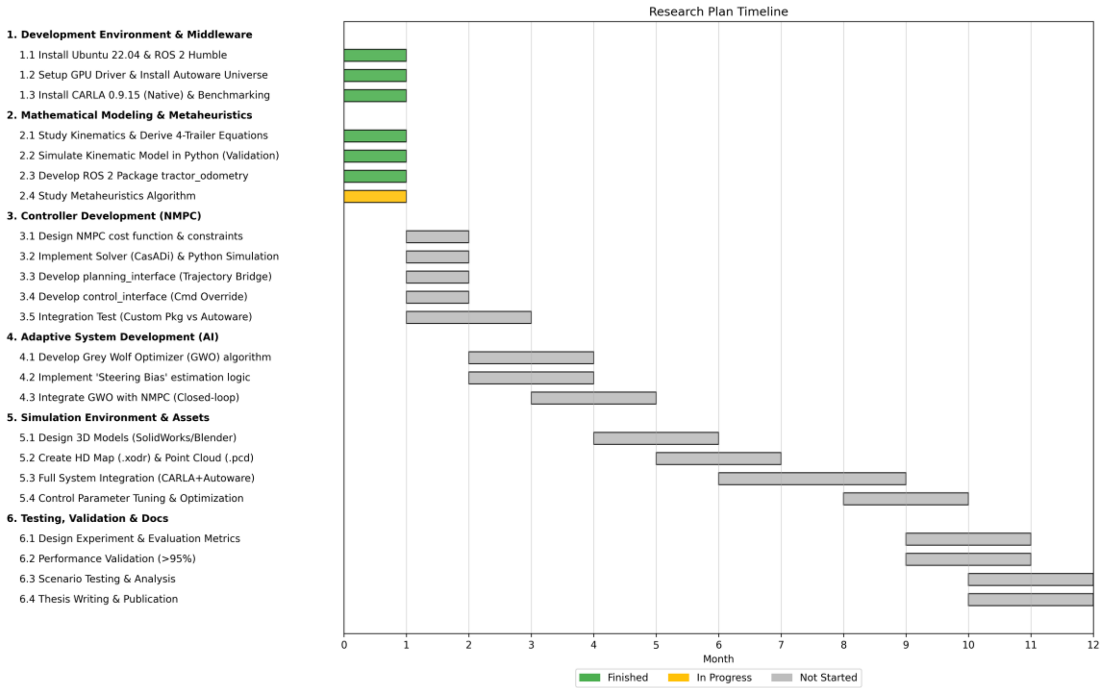

Figure 1: Research Plan Timeline. Green indicates completed tasks, Yellow indicates in-progress tasks, and Gray indicates future planned tasks.

2. Report in detail of the progress: By showing understandable and clear data including scientific data (if any) such as graph, table, and image.

2.1 Detailed Progress of Phase 1: Middleware Implementation & Performance

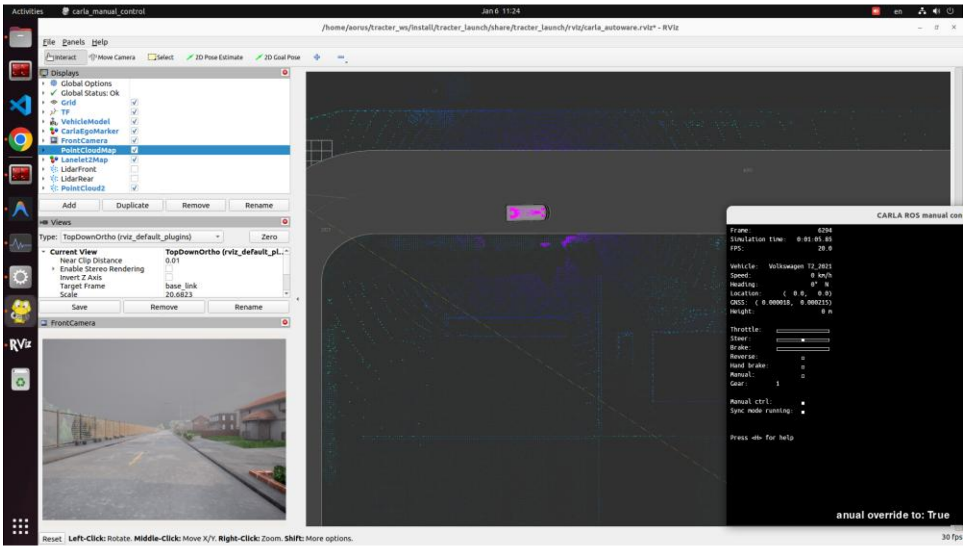

Figure 2: Implementation of CARLA, ROS 2, and Autoware bridge. The visualization shows high-fidelity 3D map data, road lanes, and real-time sensors. We successfully tested manual control via the Autoware interface.

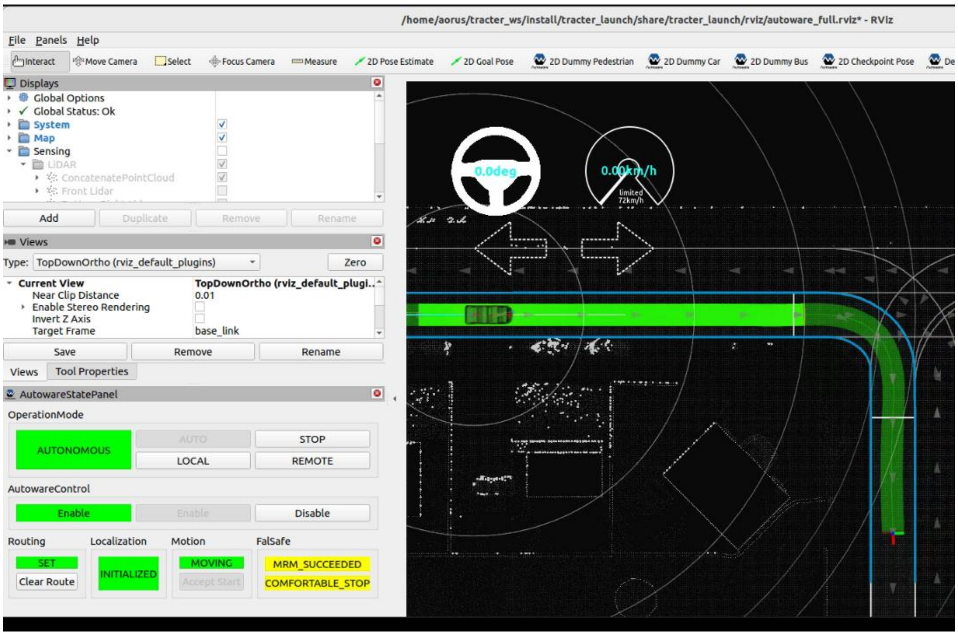

Figure 3: Integration with Autoware planning and control plugins. The system can now generate valid trajectories and execute motion commands to navigate effectively.

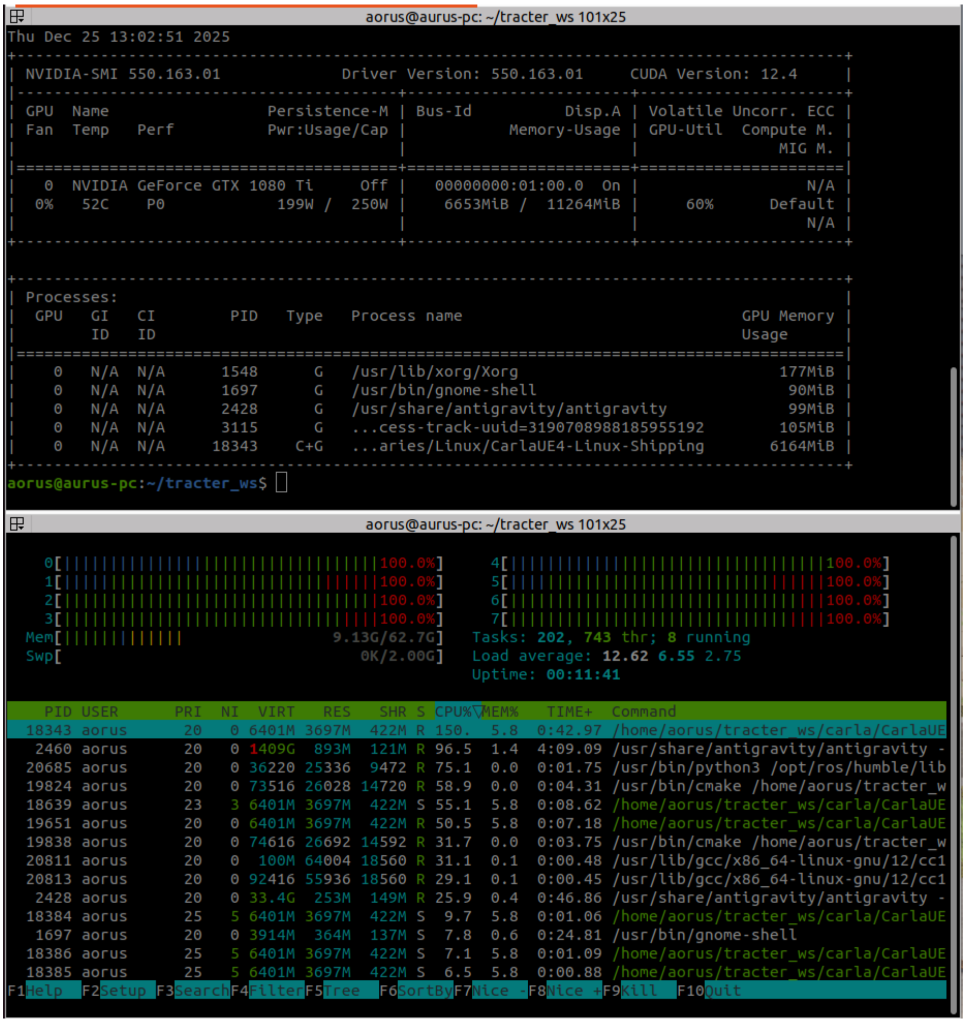

Figure 4: System resource monitoring during multi-agent simulation. The upper terminal shows heavy GPU load (60-100%), while the lower terminal confirms stable CPU usage.

2.2 Detailed Progress of Phase 2: Mathematical Modeling & Python Validation

We derived recursive kinematic equations for the multi-trailer system, capturing the complex dynamics of off-axle hitching geometry. These equations respect the non-holonomic (no-slip) constraints essential for accurate motion planning.

Package Repository: https://github.com/nontanan-linux/tracter_trailer.git

$$\dot{x}_0 = v_0 \cos\,\theta_0$$

$$\dot{y}_0 = v_0 \sin\,\theta_0$$

$$\dot{\theta}_0 = \frac{v_0}{L_0} \tan\,\delta$$

Generalized Recursive Step: k-th Trailer Unit: For any trailer unit k (where k=1,2,…,N), consisting of Drawbar k and Trailer k

1) Dolly k (P_{2k-1}): Calculated from the preceding hitch H_k.

$$P_{2k-1} = H_k - L_{bar,k} \begin{bmatrix} \cos\theta_{2k-1} \\ \sin\theta_{2k-1} \end{bmatrix}$$

2) Axle k (P_{2k}): Calculated from Dolly k.

$$P_{2k} = P_{2k-1} - L_{trl,k} \begin{bmatrix} \cos\theta_{2k} \\ \sin\theta_{2k} \end{bmatrix}$$

3) Next Hitch (H_{k+1}): Calculated from Axle k (if another trailer follows).

$$H_{k+1} = P_{2k} - d_{h,k} \begin{bmatrix} \cos\theta_{2k} \\ \sin\theta_{2k} \end{bmatrix}$$

- $L_{bar,k}$: Length of Drawbar k ($L_1, L_3, \ldots$)
- $L_{trl,k}$: Length of Trailer k ($L_2, L_4, \ldots$)
- $d_{h,k}$: Hitch offset of Trailer k ($d_{h2}, \ldots$)

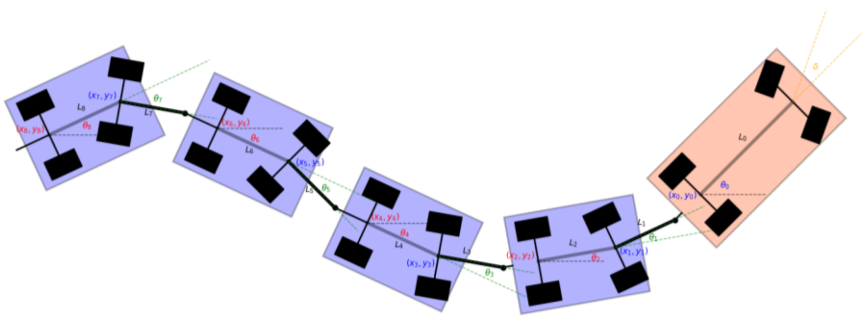

Figure 5: Kinematic diagram of the tractor with 4 drawbar trailers. The diagram defines key parameters and motion variables (theta, v). It specifically highlights the off-axle hitching distance (dh) that influences off-tracking.

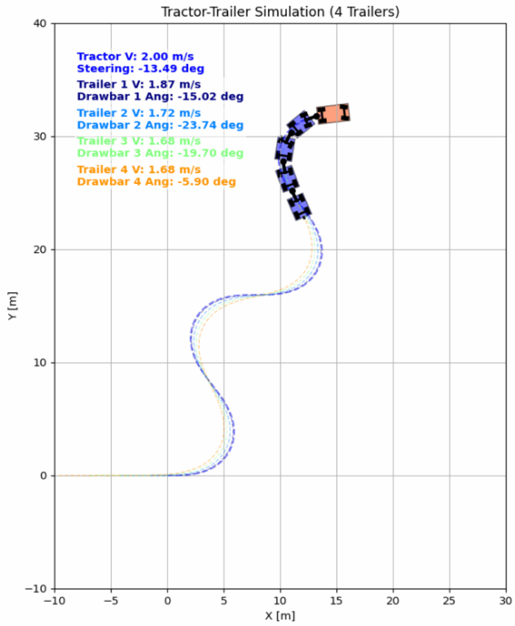

Figure 6: Python-based simulation result. The plot displays the path of the tractor (blue) and four trailers during a turn. The result confirms that our equations correctly predict the off-tracking behavior.

2.3 ROS 2 Package: tractor_odometry

We developed a ROS 2 package 'tractor_odometry' to implement these validated models. It accepts real-time steering and velocity inputs from the tractor and computes the position/velocity states for all attached trailers.

Package Repository: https://github.com/nontanan-linux/tracter_odometer.git

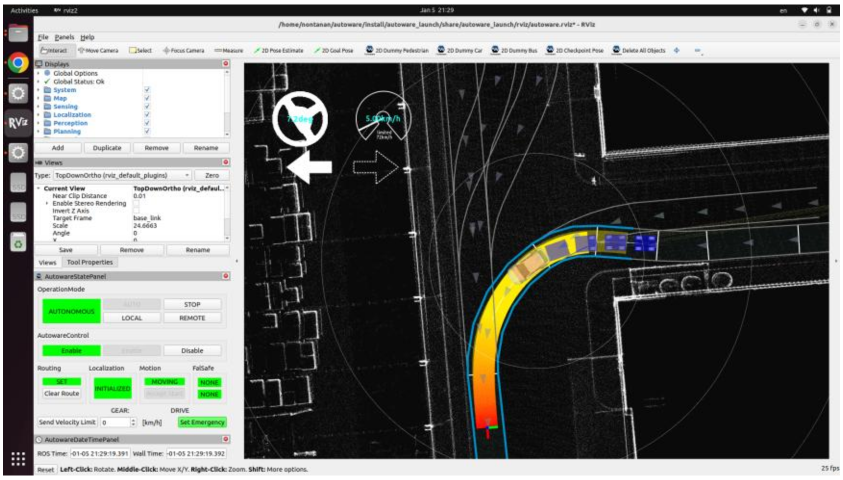

Figure 7: ROS 2 tractor_odometry visualization showing the estimated pose and velocity.

F03- Research Progress Report.

Academic works (in the period of this report) (if it is in the process of reviewing or under consideration, please explain)

○ Published in International Journal ....issue(s)

○ Published in National Journal ....issue(s)

○ Published in International Proceedings ....issue(s)

○ Published in National Proceedings ....issue(s)

○ Poster / Oral presentation in International Conference....................issue(s)

○ Poster / Oral presentation in National Conference....................issue(s)

○ Patent / Pretty Patent..........................item(s)

○ Prototype........................... item(s)

○ Award .... item(s)

○ Others.......................................

Title of academic works...

Date of presentation (if any)...

Place of presentation (if any)...

Was the work complete as planned? If there was any obstacle, please specify and explain how you adjusted the plan?

The research activities have been completed in accordance with the proposed timeline. Phase 1 (Development Environment Setup) and the critical parts of Phase 2 (Mathematical Modeling and Validation) are finished.

Obstacle & Adjustment: No significant obstacles were encountered. However, ensuring the numerical stability of the kinematic model for a 4-trailer system required more rigorous validation than initially expected. Consequently, I allocated additional time to develop a "Recursive Matrix Formulation" and a Python-based simulation (Activity 2.2) to verify the model before implementing it in ROS 2. This proactive step ensures a robust foundation for the upcoming controller development phase.

Progress of the research up to now is 25 % of the overall research plan.

(Nontanan Sommat)

12/Jan/2026

Scholarship Recipient

Part 5: For Dissertation Advisor's Comment

University Advisor's Comment

He has good performance for reaeacrh

Evaluation of student's woks ☐Excellent ☑ Good ☐Fair ☐Poor

Evaluation of scholarship ☐Should be proceeded ☐Should be suspended

(Asst. Prof. Dr. Danai Phaoharuhansa)

07/Jan/2026

University Advisor

NSTDA Advisor's Comment

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Evaluation of student's woks ☐Excellent ☐Good ☐Fair ☐Poor

Evaluation of scholarship ☐Should be proceeded ☐Should be suspended

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NSTDA Advisor